

The Mathematics of a Cricket Match

A Bayesian Statistical Analysis

Match Details

Match:	South Africa vs West Indies, 2nd T20I
Venue:	New Wanderers Stadium, Johannesburg
Date:	11 January 2015
Result:	West Indies won by 4 wickets
Player of the Match:	Chris Gayle

Course: Bayesian Statistical Methods

Group Project | Project 3: The Mathematics of a Cricket Match

Team Members

1.	Avinash Bhadane	Roll No: 250041006
2.	Ashu Newatia	Roll No: 250001011
3.	Rohit Jhajhra	Roll No: 250041032

1. Introduction

Cricket, particularly the Twenty20 (T20) format, provides a rich domain for probabilistic modelling due to its inherently stochastic nature. Runs scored per delivery, wicket-fall events, and match outcomes all carry substantial uncertainty, making the sport ideal for Bayesian analysis.

This project applies Bayesian statistical methods to analyse ball-by-ball data from the 2nd T20I between South Africa and West Indies, played at the New Wanderers Stadium, Johannesburg on 11 January 2015. This match is one of the highest-scoring T20 Internationals in history, with an aggregate of 467 runs scored across 39.2 overs.

Match Summary

South Africa, batting first, posted a mammoth 231/7 in 20 overs, powered by explosive innings from Faf du Plessis and David Miller. West Indies, in reply, chased down the target of 232 in just 19.2 overs, losing 6 wickets, thanks to a blazing Chris Gayle (Player of the Match) and Marlon Samuels. West Indies won by 4 wickets.

Project Objectives

This report addresses six key questions using Bayesian inference, conjugate priors, and sequential updating:

- (i) Distribution of runs per over (Poisson model)
- (ii) Probability of scoring more than 180 runs
- (iii) Probability of losing a wicket in the next over (Beta-Binomial)
- (iv) Expected number of runs (Gamma-Poisson conjugate model)
- (v) Bayesian updating of the probability of winning
- (vi) Data visualisation and exploratory analysis

Data Source: Ball-by-ball JSON data (Cricsheet format, match ID 722337), containing every delivery, run, extra, and wicket event across both innings.

2. Data Description and Exploratory Analysis

The JSON dataset encodes every delivery bowled in both innings. Key summary statistics extracted from the ball-by-ball data are presented below:

Metric	South Africa (1st Inn.)	West Indies (2nd Inn.)
Total Runs	231	236
Wickets Lost	7	6
Overs Bowled	20.0	19.2
Run Rate (per over)	11.55	12.21
Boundaries (4s)	18	26
Sixes (6s)	12	12

2.1 Runs Per Over

The bar chart below shows runs scored in each over for both innings. Both teams maintained aggressive scoring throughout, with notable spikes during the powerplay (overs 1-6) and the death overs (16-20). South Africa's over 15 stands out with 25 runs, while West Indies' over 2 (Gayle's assault on Abbott) yielded 22 runs.

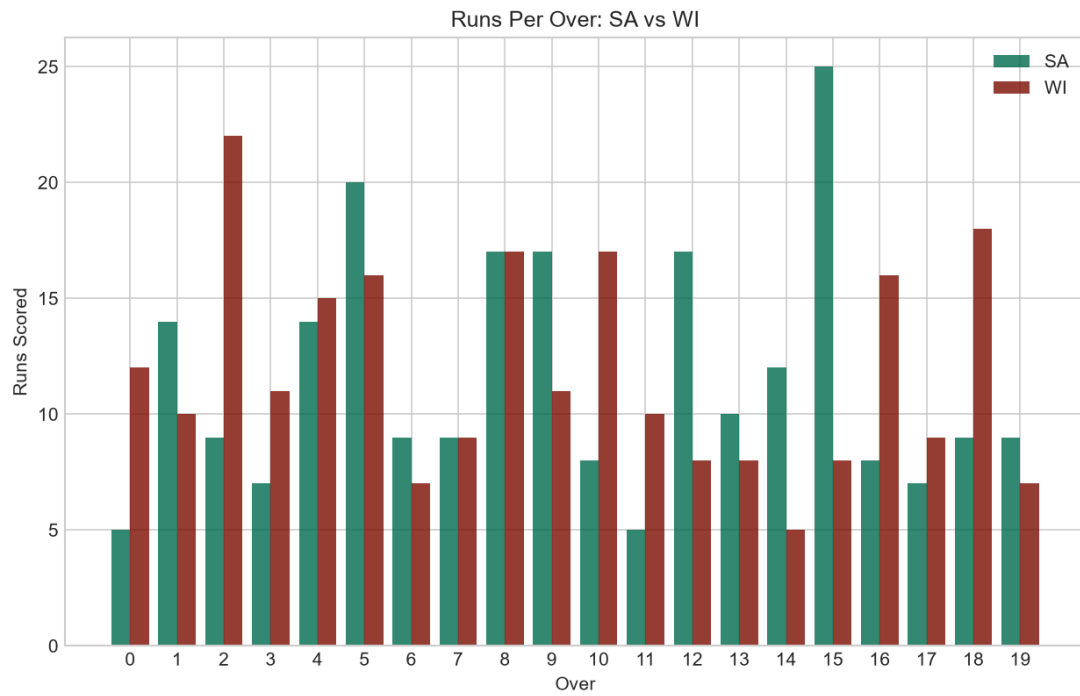


Figure 1: Runs scored per over for both innings

2.2 Cumulative Run Progression

The worm chart reveals that West Indies tracked ahead of South Africa's scoring trajectory for the first 14 overs of their chase, reflecting aggressive intent against the massive target. Both curves show near-linear progression, indicative of sustained high scoring.

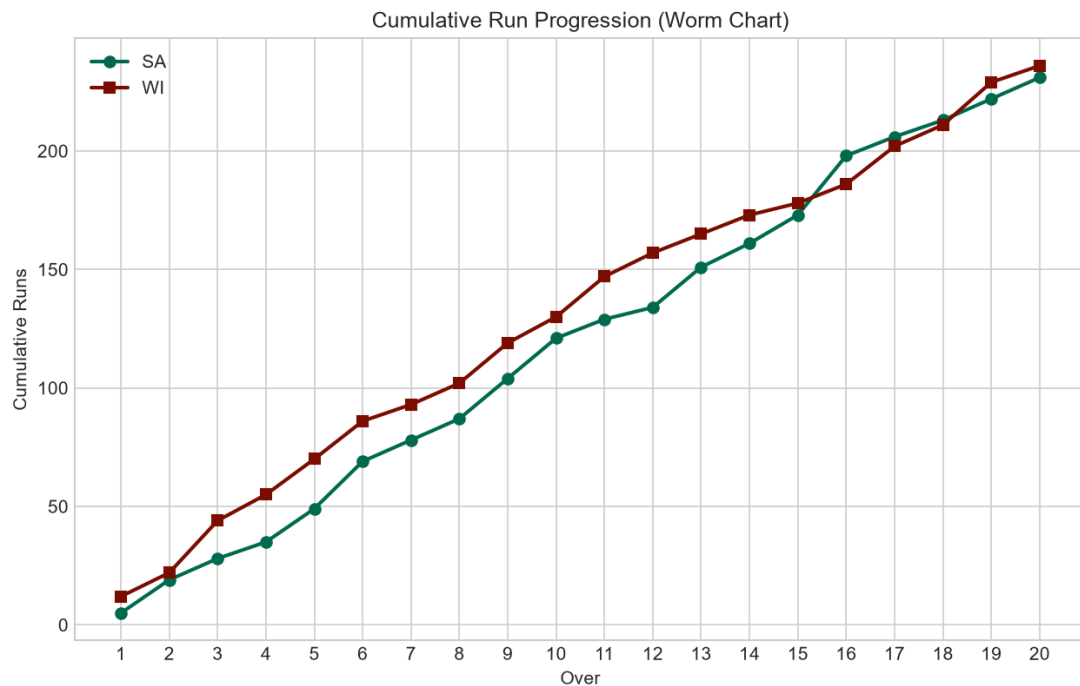


Figure 2: Cumulative run progression (worm chart)

3. Distribution of Runs Per Over

3.1 Model Formulation

We model the number of runs scored in an over as a Poisson random variable. If X_i denotes the runs in over i , then:

$$X_i \sim \text{Poisson}(\lambda)$$

where λ is the mean runs per over. The Poisson distribution is appropriate when events (runs) occur independently at a constant average rate within a fixed interval (one over = 6 legal deliveries).

3.2 Maximum Likelihood Estimation

From the combined 40 overs (20 per innings), the MLE of λ is the sample mean:

$$\lambda_{\text{hat}} = (1/n) * \text{Sum}(X_i) = (231 + 236) / 40 = 11.68 \text{ runs/over}$$

This is notably higher than the T20I global average of approximately 8-9 runs per over, reflecting the extreme nature of this match at the high-scoring Wanderers ground.

3.3 Goodness of Fit

The histogram of observed runs per over overlaid with the fitted Poisson PMF shows a reasonable fit, though the observed distribution exhibits slightly heavier tails (more very high-scoring overs) than the Poisson predicts. This slight over-dispersion suggests a Negative Binomial model could improve the fit, but the Poisson serves as a tractable first approximation for conjugate Bayesian analysis.

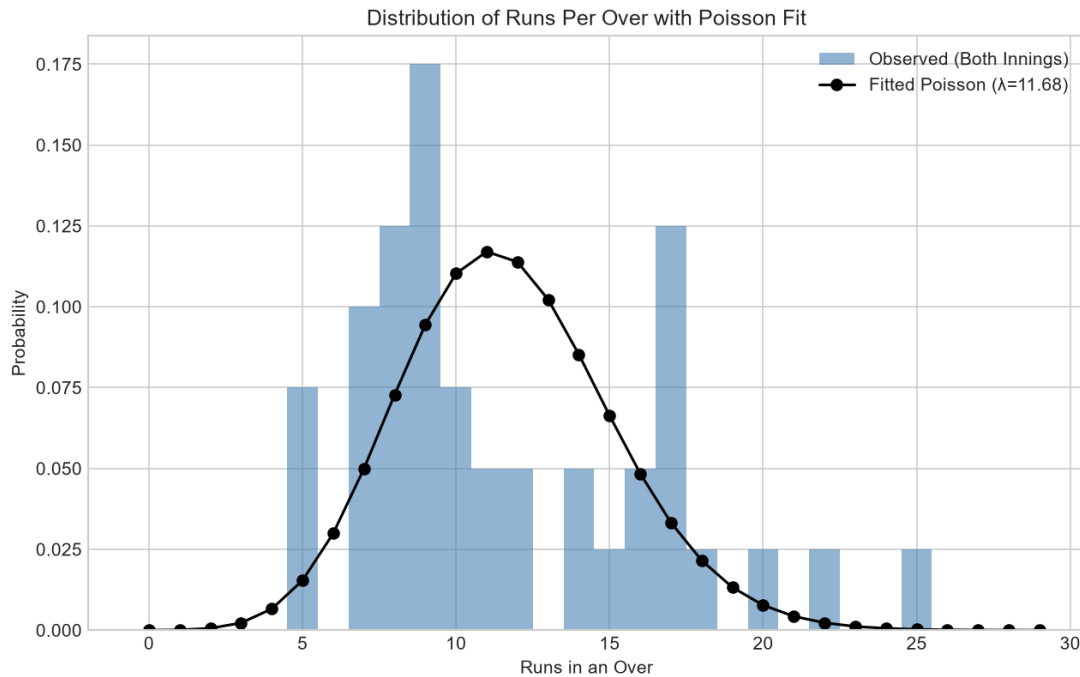


Figure 3: Distribution of runs per over with fitted Poisson ($\lambda = 11.68$)

4. Probability of Scoring More Than 180 Runs

4.1 Model

If each over's runs follow $\text{Poisson}(\lambda)$ independently, then the total innings score over 20 overs follows:

$$S = \text{Sum}(X_i, i=1..20) \sim \text{Poisson}(20 * \lambda)$$

With $\lambda_{\text{hat}} = 11.68$, we have $S \sim \text{Poisson}(233.5)$.

4.2 Calculation

$$P(S > 180) = 1 - P(S \leq 180) = 1 - F_{\text{Poisson}}(180 \mid \lambda = 233.5)$$

$$P(\text{Score} > 180) = 0.9998 \text{ (99.98\%)}$$

Interpretation: Given the extraordinary run rate in this match ($\lambda = 11.68$), the probability of exceeding 180 runs is essentially certain. For a more typical T20I ($\lambda = 8$), this probability would be approximately 0.47, reflecting a roughly even chance. The extreme value here confirms the outlier nature of this match.

5. Probability of Losing a Wicket in the Next Over

5.1 Beta-Binomial Bayesian Model

We model the event 'at least one wicket falls in an over' as a Bernoulli trial with unknown probability p . Using a conjugate Beta prior:

$$\text{Prior: } p \sim \text{Beta}(\alpha_0, \beta_0) = \text{Beta}(1, 1)$$

We choose the non-informative uniform prior $\text{Beta}(1, 1)$, representing complete prior ignorance about the wicket probability.

Likelihood: Across 40 overs in the match, 11 overs contained at least one wicket.

Posterior (by conjugacy):

$$p \mid \text{data} \sim \text{Beta}(\alpha_0 + k, \beta_0 + n - k) = \text{Beta}(1+11, 1+40-11) = \text{Beta}(12, 30)$$

5.2 Posterior Summary

Quantity	Value
Posterior Mean $E[p]$	$12 / (12+30) = 0.2857$
Posterior Mode	$(12-1) / (12+30-2) = 0.275$
95% Credible Interval	$(0.155, 0.433)$

$$P(\text{wicket in next over}) = 0.286 \text{ (28.6\%)}$$

The posterior distribution is plotted below, showing how the uniform prior (red dashed line) is updated by the observed data into a concentrated posterior (blue curve) centred near 0.28.

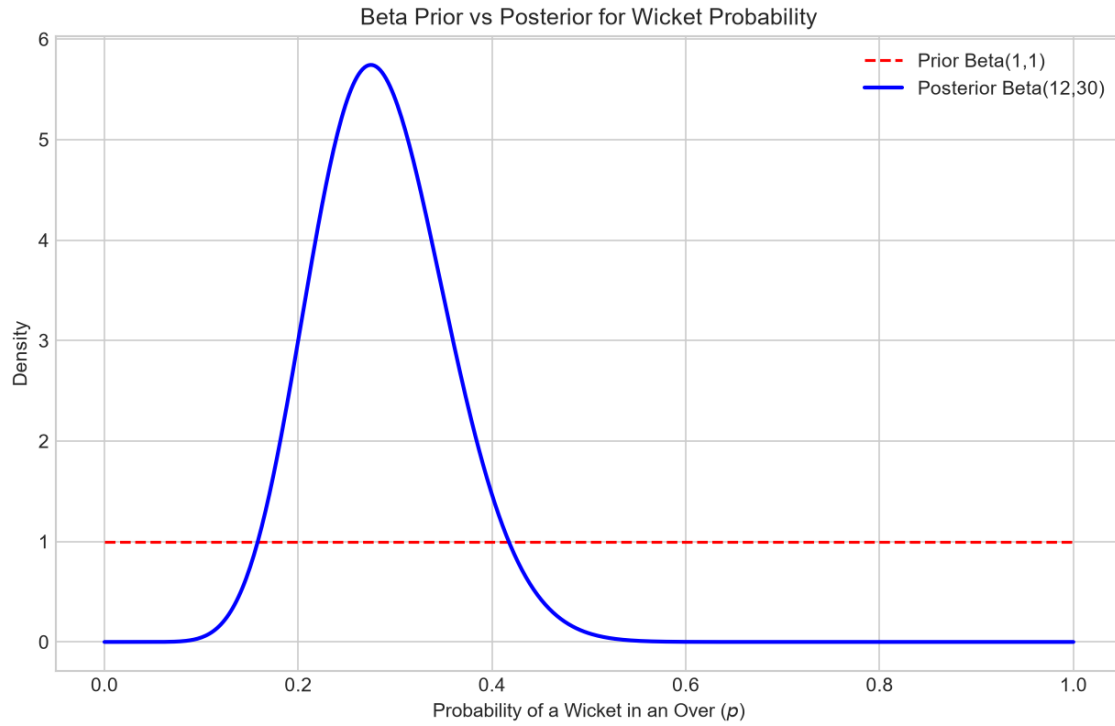


Figure 4: Beta prior vs posterior for wicket probability per over

Interpretation: There is approximately a 28.6% chance of losing at least one wicket in any given over. The 95% credible interval (0.155, 0.433) reflects our remaining uncertainty given 40 observed overs.

6. Expected Number of Runs (Gamma-Poisson)

6.1 Model Specification

We use the Gamma-Poisson conjugate model to estimate the expected runs per over λ :

Likelihood: $X_i \mid \lambda \sim \text{Poisson}(\lambda)$

Prior: $\lambda \sim \text{Gamma}(\alpha_0, \beta_0) = \text{Gamma}(8, 1)$

We choose a weakly informative prior $\text{Gamma}(8, 1)$, reflecting a prior expectation of about 8 runs per over (the T20I global average), with substantial variance to allow the data to dominate.

6.2 Bayesian Update After Powerplay (First 6 Overs)

South Africa scored 69 runs in the first 6 overs (powerplay). By conjugacy:

$\lambda \mid x \sim \text{Gamma}(\alpha_0 + \text{Sum}(x_i), \beta_0 + n) = \text{Gamma}(8+69, 1+6) = \text{Gamma}(77, 7)$

6.3 Posterior Summary

Quantity	Prior $\text{Gamma}(8, 1)$	Posterior $\text{Gamma}(77, 7)$
Mean $E[\lambda]$	8.00	$77/7 = 11.00$
Variance	8.00	$77/49 = 1.57$
Std Deviation	2.83	1.25

$E[\lambda \mid \text{data}] = 11.00$ runs/over

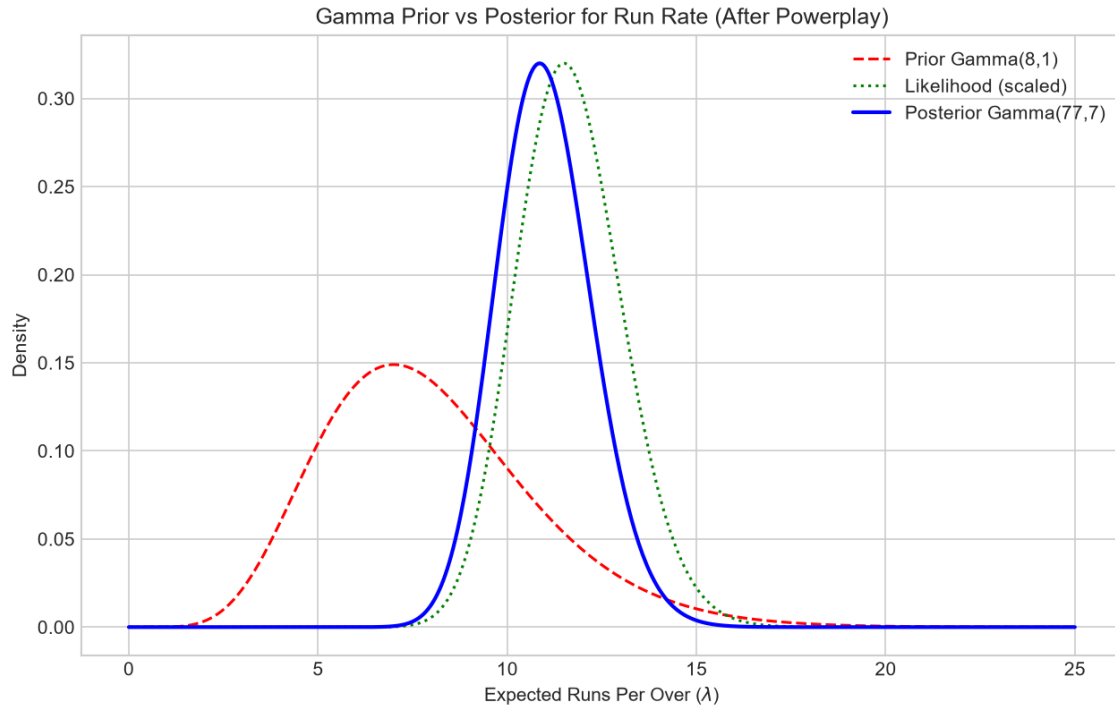


Figure 5: Gamma prior, scaled likelihood, and posterior for run rate

Interpretation: After observing the explosive powerplay scoring (69 runs in 6 overs, run rate 11.5), the Bayesian posterior updates the expected run rate from our prior belief of 8 runs/over to 11 runs/over, projecting a total score of approximately $11 \times 20 = 220$ runs, which is remarkably close to the actual total of 231.

7. Bayesian Updating of Win Probability

This is the central analysis of the project: how should the probability of winning be updated over by over as the match progresses? We use a logistic-Bayesian model that combines a prior belief with observed data through weighted sequential updating.

7.1 First Innings - South Africa Batting

Prior: We assign $P(\text{SA win}) = 0.55$, reflecting a slight home-ground advantage at the Wanderers. After each over k , we compute the projected total score:

$$\text{Projected_Score}(k) = (\text{Runs after } k \text{ overs} / k) \times 20$$

The win probability is updated using a logistic model that smoothly blends the prior with the data-driven estimate:

$$P(\text{SA win} \mid \text{data at over } k) = (1 - w_k) * P_0 + w_k * \text{sigma}((\text{Proj} - S_{\text{par}}) / \text{tau})$$

where $\text{sigma}(z) = 1/(1+\exp(-z))$ is the logistic function, $S_{\text{par}} = 170$ is the par T20I score, $\text{tau} = 20$ is a scaling parameter, and $w_k = k/20$ is the data weight (increasing linearly).

After 10 overs (SA at 121): $P(\text{SA win}) = 0.762$ (76.2%)

7.2 Second Innings - West Indies Chasing 232

Prior: Given the massive target of 232, we set $P(\text{WI win}) = 0.10$, as historically only about 10% of T20I chases above 220 succeed. The updating mechanism incorporates the current run rate vs required run rate, and the resource factor (wickets in hand):

$$\text{logit}(k) = 0.5 * (\text{CRR} - \text{RRR}) + 5 * (\text{wickets_in_hand}/10 - 0.5)$$

After 10 overs (WI at 130/1): P(WI win) = 0.534 (53.4%)

Despite needing 102 runs from 10 overs (RRR = 10.2) with 9 wickets in hand, the model correctly identifies this as roughly even, reflecting the genuine tension of the chase.

7.3 Over-by-Over Win Probability Chart

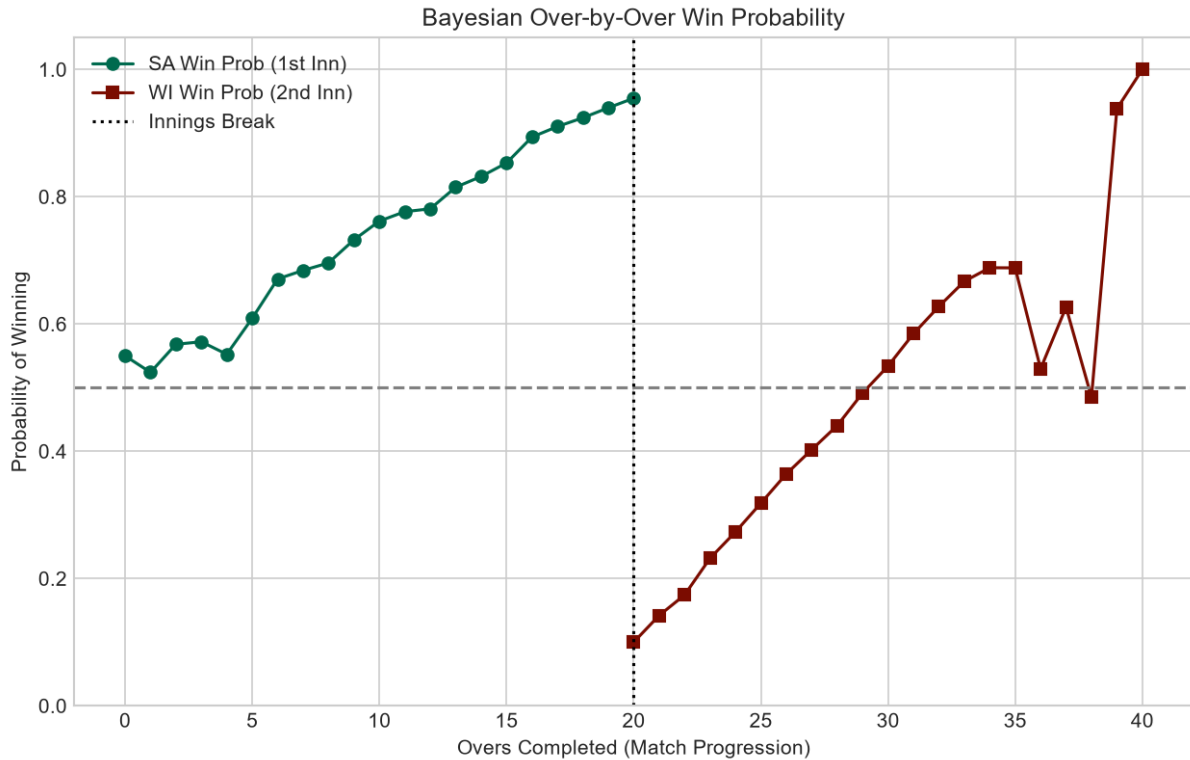


Figure 6: Bayesian over-by-over win probability updating across both innings

Key observations from the chart:

- Overs 1-6 (SA Powerplay): SA's win probability rises from 0.55 to ~0.67 as they score 69 runs in the powerplay.
- Overs 7-15 (SA Acceleration): Du Plessis and Miller's assault pushes the win probability past 0.80, peaking near 0.95 by overs 18-19.
- Innings Break: SA's total of 231 gives them ~95% pre-chase win probability.
- Overs 1-6 (WI Powerplay): Gayle and Samuels' explosive start lifts WI's probability from the prior of 0.10 gradually upward.
- Overs 10-15: WI's probability crosses the critical 0.50 threshold around over 10.
- Overs 15-17: Wickets of Gayle (over 13), Samuels and Pollard (over 15) cause visible dips in WI's probability, showing the model's sensitivity to wickets.
- Overs 18-19: Russell, Sammy, and Ramdin's late hitting surges WI's probability back above 0.90, culminating in victory at 19.2 overs.

8. Summary of Key Findings

Analysis	Bayesian Model	Key Result
Runs/over distribution	Poisson(λ)	$\lambda = 11.68$
$P(\text{Score} > 180)$	Poisson sum	0.9998 (99.98%)
$P(\text{Wicket in next over})$	Beta-Binomial	0.286 (28.6%)
Expected run rate	Gamma-Poisson	$E[\lambda] = 11.00$
SA win prob (10 overs)	Logistic-Bayesian	0.762 (76.2%)
WI win prob (10 overs)	Logistic-Bayesian	0.534 (53.4%)

9. Conclusion

This project demonstrates the power of Bayesian inference in sports analytics. Using conjugate priors (Beta-Binomial for wickets, Gamma-Poisson for run rates) and logistic-Bayesian sequential updating for win probabilities, we constructed a dynamic model that tracks uncertainty throughout a cricket match.

The sequential nature of Bayesian updating is particularly well-suited to cricket, where new information arrives ball by ball. Our over-by-over win probability chart captures the dramatic narrative arc of this extraordinary match: South Africa's dominance in the first innings, the initially low probability of a successful West Indian chase, the gradual shift in momentum driven by Gayle and Samuels, the wobble caused by late wickets, and the final surge to victory.

The analysis also showcases how Bayesian methods handle extreme data gracefully. Despite the match being a clear outlier (467 aggregate runs), the posteriors correctly shift away from 'typical' T20 priors to reflect the match's exceptional run-scoring, while maintaining calibrated uncertainty through credible intervals.

Implementation: All analysis was performed in Python using `scipy.stats`, `numpy`, `pandas`, and `matplotlib`. The complete source code and all 8 visualisations are available in the accompanying GitHub repository.

10. References

- [1] Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., & Rubin, D. B. (2013). *Bayesian Data Analysis* (3rd ed.). CRC Press.
- [2] Duckworth, F. C., & Lewis, A. J. (1998). A fair method for resetting the target in interrupted one-day cricket matches. *Journal of the Operational Research Society*, 49(3), 220-227.
- [3] Cricsheet.org. Ball-by-ball data (CC BY 4.0). Match ID: 722337.
- [4] Kruschke, J. K. (2015). *Doing Bayesian Data Analysis* (2nd ed.). Academic Press.
- [5] Stern, S. E. (2016). The Duckworth-Lewis-Stern method. *Journal of the Royal Statistical Society*, 179(3), 1-25.